

SHEAR LOCKING

SHEAR LOCKING PHENOMENON AND SOME REMEDIAL MEARSURES

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Outline



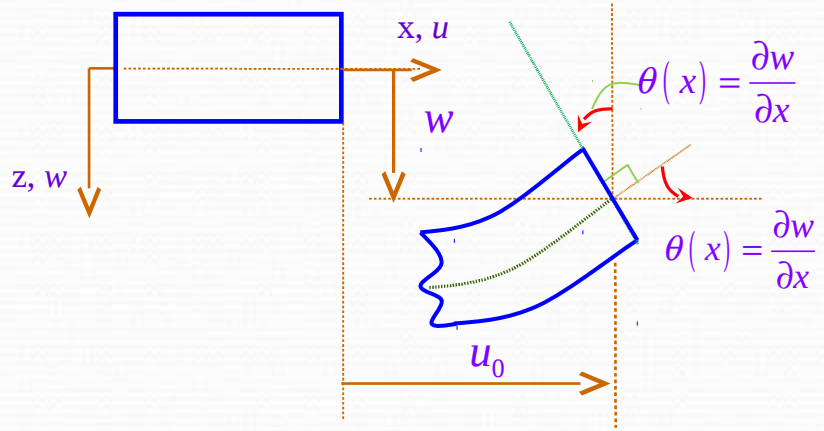
- Degenerated beam elements
- Shear locking phenomenon and remedical measures for beam
- Shear locking phenomenon and remedical measures for plates



Degenerated beam elements

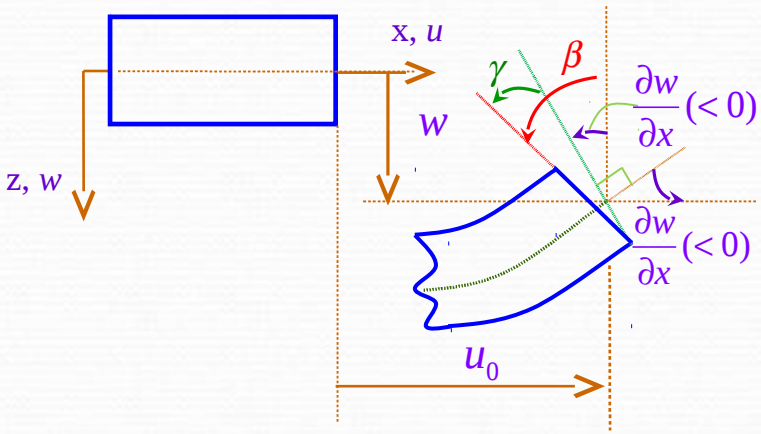
General beam elements

Euler – Bernoulli beam



- Effect of shear deformation neglected
- Basis variable: w
- Working parameters: w & $\theta=\frac{\partial w}{\partial x}$

Timoshenko beam



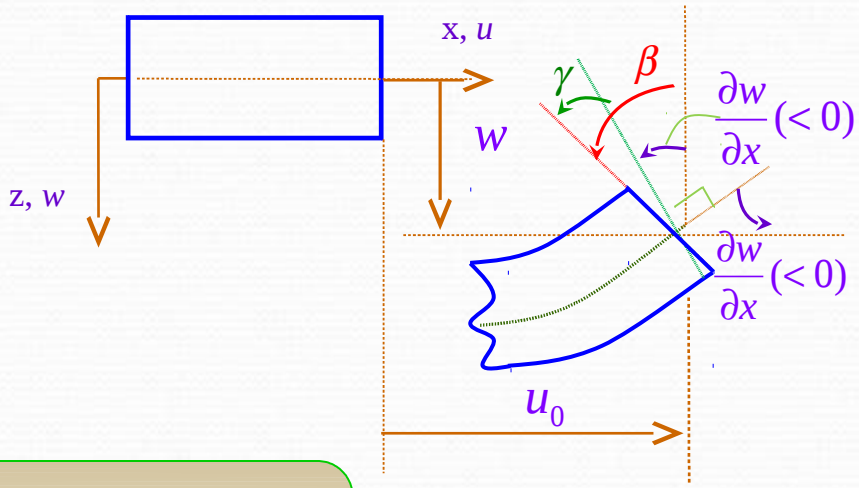
- Having effect of shear deformation
- Basis variable: w & $\beta=-\frac{\partial w}{\partial x}+\gamma$
- Working parameters: w & β

Degenerated beam elements

Timoshenko beam

Displacement formulation

$$w = w(x) \;\; ; \;\; u(x,z) = \beta(x) z$$



Strain formulation

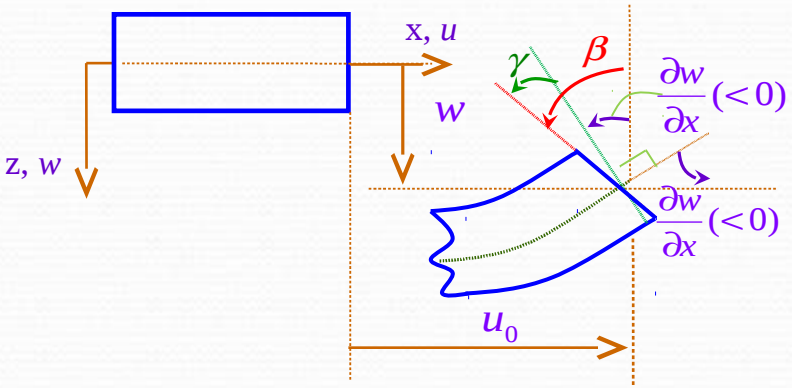
$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_z \\ \gamma_{xz} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial w}{\partial z} \\ \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial \beta(x)}{\partial x} z \\ 0 \\ \frac{\partial w(x)}{\partial x} + \beta(x) \\ \gamma \end{Bmatrix} \Rightarrow \boldsymbol{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \gamma_{xz} \end{Bmatrix} = \begin{Bmatrix} \frac{\partial \beta(x)}{\partial x} z \\ \gamma \end{Bmatrix}$$

Degenerated beam elements

Timoshenko beam

Virtual displacement principle

$$\int_V \delta\boldsymbol{\varepsilon}^T \boldsymbol{\sigma} \mathrm{d}V = \int_V \delta\mathbf{u}^T \mathbf{f} \mathrm{d}V$$



$$\mathrm{LHS} = \int_V \underbrace{\left[\frac{\partial \delta\beta}{\partial x} z \quad \delta\gamma\right]}_{\delta\boldsymbol{\varepsilon}^T} \underbrace{\begin{bmatrix} E & 0 \\ 0 & G \end{bmatrix}}_{\boldsymbol{\sigma}=\mathbf{D}\boldsymbol{\varepsilon}} \begin{Bmatrix} \frac{\partial \beta}{\partial x} z \\ \gamma \end{Bmatrix} \mathrm{d}V \qquad \beta_1$$

$$= E\int_V \frac{\partial \delta\beta}{\partial x}.\frac{\partial \beta}{\partial x} z^2 \mathrm{d}V + G\int_V \delta\gamma.\gamma \mathrm{d}V$$

$$= E\int_A z^2 \mathrm{d}A \int_0^L \frac{\partial \delta\beta}{\partial x}.\frac{\partial \beta}{\partial x} \mathrm{d}x + kGA\int_0^L \delta\gamma.\gamma \mathrm{d}x$$

$$= EI\int_0^L \frac{\partial \delta\beta}{\partial x}.\frac{\partial \beta}{\partial x} dx + kGA\int_0^L \underbrace{\left(\frac{\partial \delta w}{\partial x} + \delta\beta\right)}_{\delta\gamma} \underbrace{\left(\frac{\partial w}{\partial x} + \beta\right)}_{\gamma} \mathrm{d}x$$

$$\mathrm{RHS} = \int_V \delta\mathbf{u}^T \mathbf{f} \mathrm{d}V$$

$$= \int_V \underbrace{[\delta w \quad \delta\beta]}_{\delta\mathbf{u}^T} \underbrace{\begin{Bmatrix} f_z(x) \\ m_y(x) \end{Bmatrix}}_{\mathbf{f}} \mathrm{d}V$$

$$= \int_0^L \delta w \underbrace{p_z(x)}_{f_z(x)A} \mathrm{d}x + \int_0^L \delta\beta \underbrace{m_y(x)}_{m_V(x)A} \mathrm{d}x$$

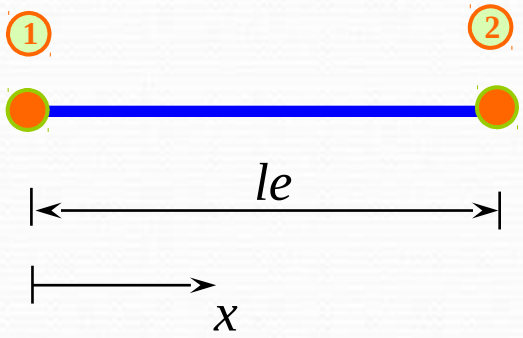
Degenerated beam elements

Timoshenko beam

Displacement interpolation

$$w = \underbrace{\begin{bmatrix} N_1 & N_2 \end{bmatrix}}_{\mathbf{N}} \underbrace{\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}}_{\mathbf{w}} \quad \Rightarrow \quad \frac{\partial w}{\partial x} = \underbrace{\frac{\partial \mathbf{N}}{\partial x}}_{\mathbf{B}} \mathbf{w} = \mathbf{B}\mathbf{w}$$

$$\beta = \underbrace{\begin{bmatrix} N_1 & N_2 \end{bmatrix}}_{\mathbf{N}} \underbrace{\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}}_{\boldsymbol{\beta}} \quad \Rightarrow \quad \frac{\partial \beta}{\partial x} = \underbrace{\frac{\partial \mathbf{N}}{\partial x}}_{\mathbf{B}} \boldsymbol{\beta} = \mathbf{B}\boldsymbol{\beta}$$



$$N_1(x) = 1 - \frac{x}{l^e} \quad ; \quad N_2(x) = \frac{x}{l^e}$$

General displacement interpolation

$$\begin{bmatrix} w \\ \beta \end{bmatrix} = \begin{bmatrix} N_1 & 0 & N_2 & 0 & \mathsf{L} \\ 0 & N_1 & 0 & N_2 & \mathsf{L} \end{bmatrix} \begin{bmatrix} w_1 \\ \beta_1 \\ w_2 \\ \beta_2 \\ \mathsf{M} \end{bmatrix}$$

Degenerated beam elements



- ***Timoshenko beam***



FEM equation

$$\text{LHS} = EI\int_0^L \frac{\partial\delta\beta}{\partial x}\cdot\frac{\partial\beta}{\partial x}dx + kGA\int_0^L \underbrace{\left(\frac{\partial\delta w}{\partial x}+\delta\beta\right)}_{\delta\gamma}\underbrace{\left(\frac{\partial w}{\partial x}+\beta\right)}_{\gamma}\mathrm{d}x$$

$$= EI\,\delta\mathbf{U}^T\int_0^L \mathbf{B}^T\mathbf{B}dx\mathbf{U} + kGA\,\delta\mathbf{U}^T\int_0^L \left(\mathbf{B}+\mathbf{N}\right)^T\left(\mathbf{B}+\mathbf{N}\right)^T\mathrm{d}x\mathbf{U}$$

$$\text{RHS} = \int_0^L \delta w p(x)\mathrm{d}x + \int_0^L \delta\beta m(x)\mathrm{d}x$$

$$= \delta\mathbf{U}^T\int_0^L \mathbf{N}^T p(x)\mathrm{d}x + \delta\mathbf{U}^T\int_0^L \mathbf{N}^T m(x)\mathrm{d}x$$

$$\mathbf{KU} = \mathbf{F}$$

where

$$\mathbf{K} = EI\int_0^L \mathbf{B}^T\mathbf{B}dx + kGA\int_0^L \left(\mathbf{B}+\mathbf{N}\right)^T\left(\mathbf{B}+\mathbf{N}\right)^T\mathrm{d}x$$

$$\mathbf{F} = \int_0^L \mathbf{N}^T p(x)\mathrm{d}x + \int_0^L \mathbf{N}^T m(x)\mathrm{d}x$$



Shear locking and remedical measures

Shear locking phenomenon

Stiffness matrix

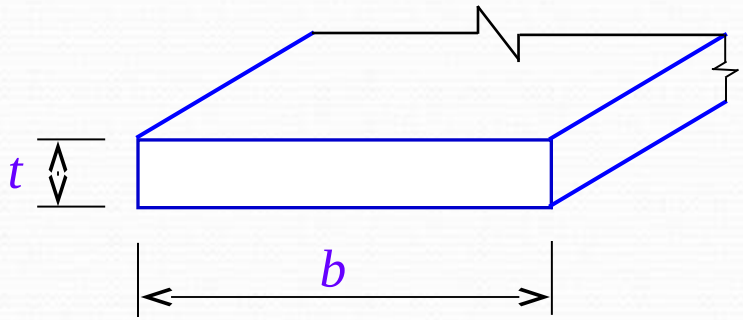
$$\mathbf{K} = EI\int_0^L \mathbf{B}^T\mathbf{B}dx + kGA\int_0^L \left(\mathbf{B}+\mathbf{N}\right)^T\left(\mathbf{B}+\mathbf{N}\right)^T dx$$

$$\frac{\mathbf{K}}{EI} = \int_0^L \mathbf{B}^T\mathbf{B}dx + \alpha\int_0^L \left(\mathbf{B}+\mathbf{N}\right)^T\left(\mathbf{B}+\mathbf{N}\right)^T dx$$

$$\alpha = \frac{kGA}{EI} = \frac{kGA}{E\dfrac{At^2}{12}} = \frac{12kG}{Et^2}$$

When $t \to 0$, $\alpha \to$ very large $\Rightarrow \mathbf{K}$ very large

Shear locking



Cross section of beam

The element is too stiff

w is too small

Result converge very slowly

Shear locking and remedical measures

Shear locking phenomenon example

Linear element



$$w=\left(1-\frac{x}{l^e}\right)w_1+\frac{x}{l^e}w_2 \quad \Rightarrow \quad \frac{\partial w}{\partial x}=\frac{w_2-w_1}{l^e}=\text{constant}$$

$$\beta=\left(1-\frac{x}{l^e}\right)\beta_1+\frac{x}{l^e}\beta_2 \ , \ l^e \text{ length of element}$$

For thin beam $\gamma \to 0$, $\beta=\frac{\partial w}{\partial x} \Leftrightarrow \left(\frac{\beta_2}{l^e}-\frac{\beta_1}{l^e}\right)x+\beta_1=\frac{w_2-w_1}{l^e}$

$$\beta_2-\beta_1=0$$

$$\beta_1=\frac{w_2-w_1}{l^e}$$

$$\beta=\left(1-\frac{x}{l^e}\right)\beta_1+\frac{x}{l^e}\beta_2=\beta_1=\text{constant} \Rightarrow EI\int_0^L \underbrace{\left(\frac{\partial \beta}{\partial x}\right)^2}_{\mathbf{B}^T\mathbf{B}} dx=0 \text{ (bending)}$$

Bending energy contribution vanish ⇒ Shear locking

Shear locking and remedical measures

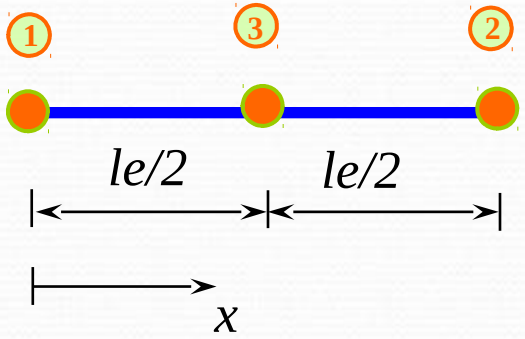
Remedical measures

Consistent interpolation — Reduced integration — Assumed strain field

Consistent interpolation

$$w=\left(1-\frac{x}{l^e}\right)\left(1-\frac{2x}{l^e}\right)w_1-\frac{x}{l^e}\left(1-\frac{2x}{l^e}\right)w_2+\frac{4x}{l^e}\left(1-\frac{x}{l^e}\right)w_3$$

$$\beta=\left(1-\frac{x}{l^e}\right)\beta_1+\frac{x}{l^e}\beta_2$$



For thin beam $\gamma \overset{should}{\rightarrow} 0$, $\beta \approx \frac{\partial w}{\partial x} = f(x)$

Bending function $EI\int_0^L \underbrace{\left(\frac{\partial \beta}{\partial x}\right)^2}_{\mathbf{B}^T\mathbf{B}} dx \neq 0$

Bending action can contribute » No shear locking!!!

Shear locking and remedical measures

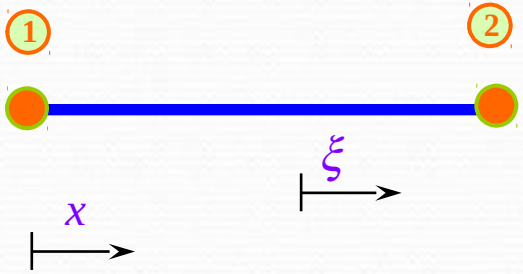
Remedical measures

Reduced integration



$$w=\left(1-\frac{x}{l^e}\right)\underbrace{w_1}_{0(clamp)}+\frac{x}{l^e}w_2=\frac{x}{l^e}w_2 \qquad w=\frac{1}{2}(1+\xi)\,w_2$$

$$\beta=\left(1-\frac{x}{l^e}\right)\underbrace{\beta_1}_{0(clamp)}+\frac{x}{l^e}\beta_2=\frac{x}{l^e}\beta_2 \qquad \beta=\frac{1}{2}(1+\xi)\,\beta_2$$



2 GP

$$\gamma=\frac{\partial w}{\partial x}+\beta\to 0,\ \frac{w_2}{l^e}-\frac{x}{l^e}\beta_2=0$$

$$\begin{bmatrix}\frac{1}{l^e} & \frac{-1}{2}\left(1+\frac{1}{\sqrt{3}}\right)\\ \frac{1}{l^e} & \frac{-1}{2}\left(1-\frac{1}{\sqrt{3}}\right)\end{bmatrix}\begin{bmatrix}w_2\\ \beta_2\end{bmatrix}=\begin{Bmatrix}0\\ 0\end{Bmatrix}$$

$w_2=0\ ;\ \beta_2=0$

Beam is modelled by a 2 node beam element

If 1 GP $x=l/2\ ;\ \xi=0$

$$\gamma=\frac{\partial w}{\partial x}+\beta\to 0,\ \frac{w_2}{l^e}-\frac{x}{l^e}\beta_2=0\Rightarrow w_2=\frac{\beta_2 l^e}{2}$$

Have chance!!!

Shear locking and remedical measures



➢ ***Remedical measures***



Assumed strain field

- Assumed shear strain variation to be one oder lower than those of w and β

$$w=\sum_{i=1}^{n} N_i w_i \quad ; \quad \beta=\sum_{i=1}^{n} N_i \beta_i \quad ; \quad \gamma=\sum_{i=1}^{n-1} N_i^{*} \gamma_i$$

Elements	w	β	γ
2-node	linear	linear	constant
3-node	quadratic	quadratic	linear



Shear locking and remedical measures

➢ *Notes*



1. Effect of shear locking reduces when either higher order elements are used or mesh is more refined

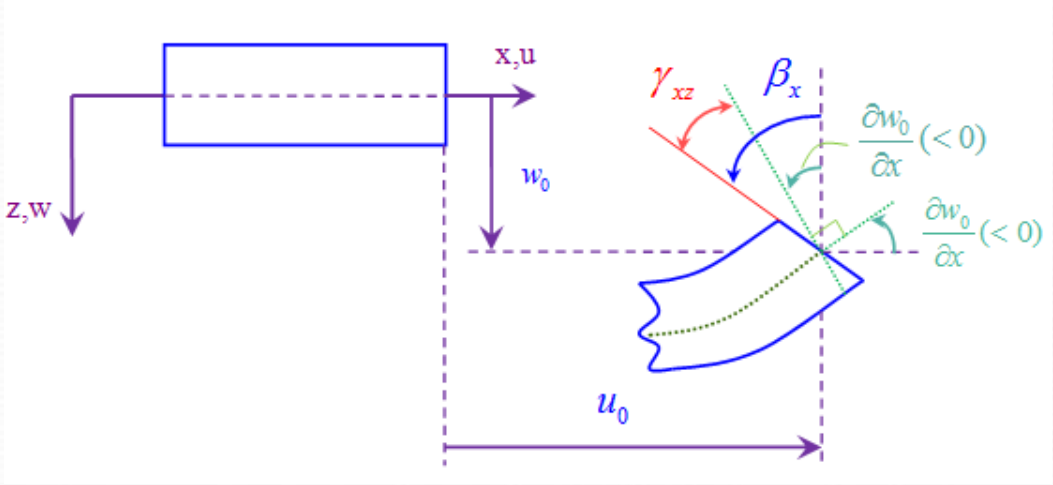
2. Selective integration is computationally more efficient but spurious zeors energy mode may be introduced

3. Assumed strain field is most rational. Supporting theories and proof are readily available and reliable elements have been developed



Shear locking and remedical measures

- *Remedical measure for plates*

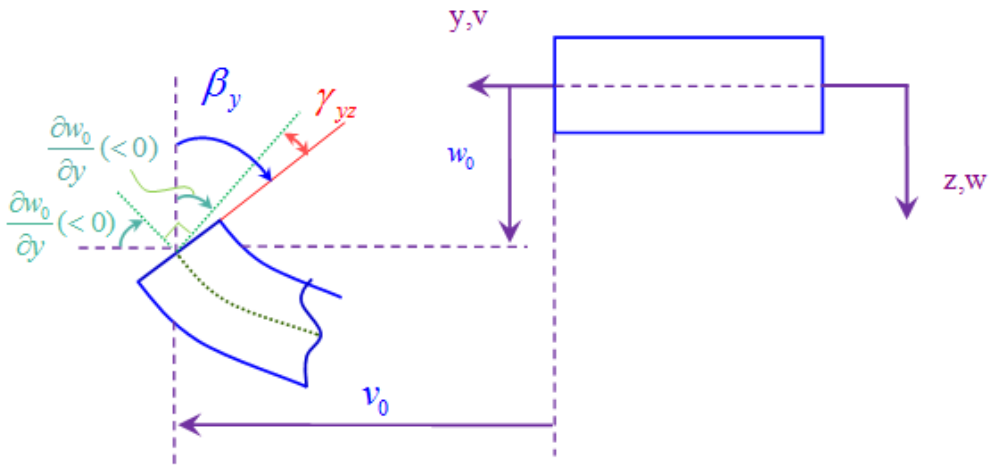


x axis

$$\gamma_{xz} = \frac{\partial w_0}{\partial x} + \beta_x = w'_x + \beta_x$$

y axis

$$\gamma_{yz} = \frac{\partial w_0}{\partial y} + \beta_y = w'_y + \beta_y$$

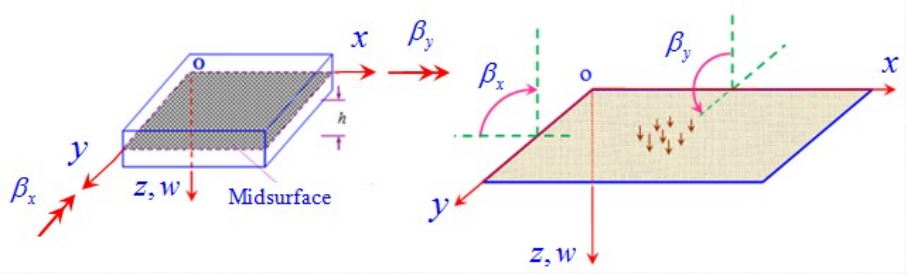


Shear locking and remedical measures

Remedical measure for plates

Assumed strain field

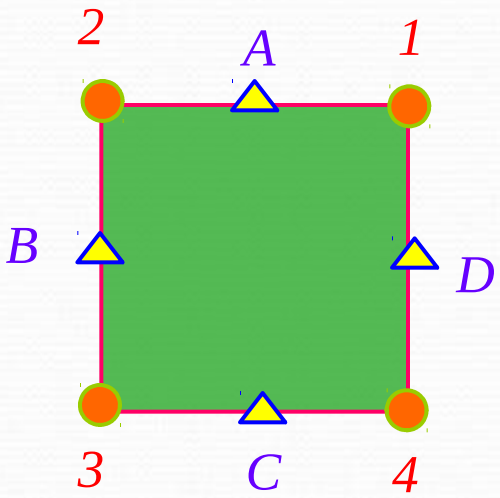
$$w=\sum_{i=1}^{n} N_i w_i \quad ; \quad \beta_x=\sum_{i=1}^{n} N_i \beta_{xi} \quad ; \quad \beta_y=\sum_{i=1}^{n} N_i \beta_{yi}$$



$$\gamma_{xz}=\sum_{i=1}^{n-1} N_i^* \gamma_{xzi} \quad ; \quad \gamma_{yz}=\sum_{i=1}^{n-1} N_i^* \gamma_{yzi}$$

$$\gamma_{yz}=\left(w'_y+\beta_y\right)\Big|_s \; ; \; \gamma_{xz}=\left(w'_x+\beta_x\right)\Big|_s$$

- Applied for Q4 (MITC4)



$$\gamma_{xz}=\sum_{i=1}^{n-1} N_i^* \gamma_{xzi}=\frac{1}{2}(1+\eta)\,\gamma_{xz}^{A}+\frac{1}{2}(1-\eta)\,\gamma_{xz}^{C}$$

N_i^*

$$\gamma_{yz}=\sum_{i=1}^{n-1} N_i^* \gamma_{yzi}=\frac{1}{2}(1+\xi)\,\gamma_{yz}^{D}+\frac{1}{2}(1-\xi)\,\gamma_{yz}^{B}$$

Shear locking and remedical measures

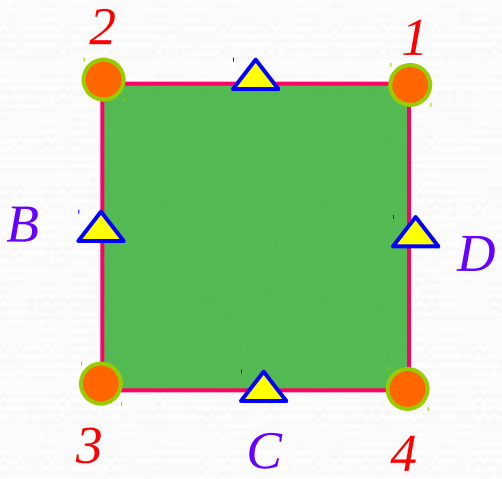
Remedical measure for plates

Assumed strain field

$$\gamma_{xz}^{A}=\left(\frac{w_1-w_2}{a}\right)\vec{i}+\left(\frac{\beta_{x1}+\beta_{x2}}{2}\right)\vec{j}\quad;\quad \gamma_{xz}^{C}=\left(\frac{w_4-w_3}{a}\right)\vec{i}+\left(\frac{\beta_{x4}+\beta_{x3}}{2}\right)\vec{j}$$

w' β

$$\gamma_{yz}^{D}=\left(\frac{w_1-w_4}{b}\right)\vec{i}-\left(\frac{\beta_{y1}+\beta_{y4}}{2}\right)\vec{j}\quad;\quad \gamma_{yz}^{B}=\left(\frac{w_2-w_3}{b}\right)\vec{i}-\left(\frac{\beta_{y2}+\beta_{y3}}{2}\right)\vec{j}$$



$$\boldsymbol{\gamma}=\mathbf{N}_\gamma\,\boldsymbol{\gamma}\big|_s=\underbrace{\mathbf{N}_\gamma\mathbf{B}_\gamma}_{\mathbf{B}_s}\mathbf{U}=\mathbf{B}_s\mathbf{U}$$

$$\mathbf{N}_\gamma=\begin{bmatrix}\frac{1}{2}(1+\eta) & \frac{1}{2}(1-\eta) & 0 & 0\\ 0 & 0 & \frac{1}{2}(1+\xi) & \frac{1}{2}(1-\xi)\end{bmatrix}$$

$$\mathbf{B}_\gamma=\begin{bmatrix}\frac{1}{a} & 0 & \frac{1}{2} & -\frac{1}{a} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{a} & 0 & \frac{1}{2} & \frac{1}{a} & 0 & \frac{1}{2}\\ \frac{1}{b} & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{b} & -\frac{1}{2} & 0\\ 0 & 0 & 0 & \frac{1}{b} & -\frac{1}{2} & 0 & -\frac{1}{b} & -\frac{1}{2} & 0 & 0 & 0 & 0\end{bmatrix}$$

THANKS YOUR ATTENTION

